Lecture Notes
EC226 Term 2 2024/25

Stationary and Non-Stationary Processes

Readings:

1. Stock and Watson (2003): Chapter 14
2. Dougherty (2016): Chapter 13
3. Wooldridge (2013): Chapter 18

Contents

1 Definition of (weak) stationarity

A series, y_t, is said to be weakly stationary if:

(i) $E(y_t) = \mu, \quad \forall \ t = 1, \ldots, T$

(ii) $V(y_t) = E\left[(y_t - \mu)^2\right] = \sigma_y^2 \quad \forall \ t = 1, \ldots, T$

(iii) $cov(y_t, y_{t-h}) = E\left[(y_t - \mu)(y_{t-h} - \mu)\right] = \gamma_h \quad \forall \ t = 1, \ldots, T$

Consider the simple AR(1) model:

$$y_t = \mu + \phi y_{t-1} + \varepsilon_t \text{ where } \phi < 1 \tag{1}$$

lagging equation (1) by one period we get

$$y_{t-1} = \mu + \phi y_{t-2} + \varepsilon_{t-1} \Rightarrow y_t = \mu + \phi(\mu + \phi y_{t-2} + \varepsilon_{t-1}) + \varepsilon_t$$

and substituting this equation into equation (1), we get

$$y_t = \mu(1 + \phi) + \phi^2 y_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t \tag{2}$$

Now lagging equation (1) by two periods and substituting this equation for y_{t-2} into equation (2) we get:

$$\begin{aligned} y_t &= \mu(1 + \phi) + \phi^2(\mu + \phi y_{t-3} + \varepsilon_{t-2}) + \phi\varepsilon_{t-1} + \varepsilon_t \\ &= \mu(1 + \phi + \phi^2) + \phi^3 y_{t-3} + \phi^2\varepsilon_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t \end{aligned}$$

substituting out y_{t-3} from the above equation we get

$$\begin{aligned} y_t &= \mu(1 + \phi + \phi^2) + \phi^3(\mu + \phi y_{t-4} + \varepsilon_{t-3}) + \phi^2\varepsilon_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t \\ &= \mu(1 + \phi + \phi^2 + \phi^3) + \phi^4 y_{t-4} + \phi^3\varepsilon_{t-3} + \phi^2\varepsilon_{t-2} + \phi\varepsilon_{t-1} + \varepsilon_t \end{aligned} \tag{3}$$

Continuing to back substitute, we will eventually get:

$$y_t = \mu(1 + \phi + \phi^2 + \phi^3 + \ldots) + \phi^t y_0 + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}$$

If $|\phi| < 1$ the 1st term on the right hand side of the above equation can be written as

$$\mu(1 + \phi + \phi^2 + \phi^3 + \ldots) = \frac{\mu}{(1 - \phi)}$$

as this is a (near) infinite sum of a geometric series and hence we have

$$y_t = \frac{\mu}{1 - \phi} + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j} \tag{4}$$

Taking expectations of equation (4) we have

$$E(y_t) = \frac{\mu}{1-\phi} + E\left(\sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}\right) = \frac{\mu}{1-\phi} + \sum_{j=0}^{t-1} \phi^j E(\varepsilon_{t-j}) = \frac{\mu}{1-\phi} \tag{5}$$

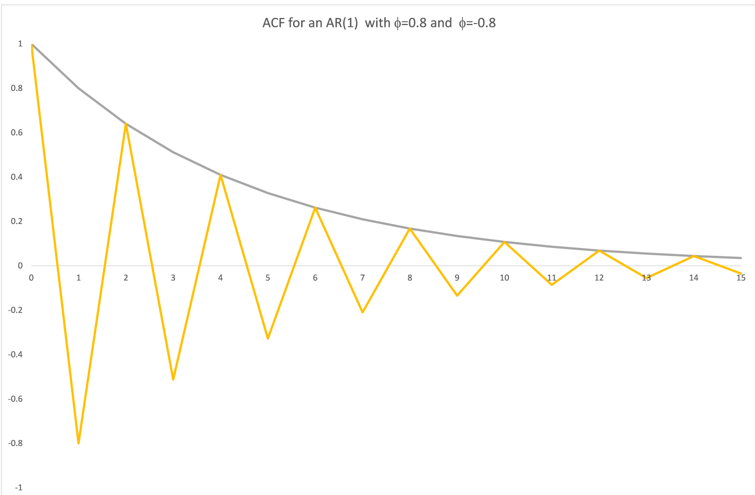
which satisfies condition (i) of weak stationarity.
The variance of equation (4)

$$V(y_t) = V\left(\frac{\mu}{1-\phi} + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}\right) = V\left(\sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}\right) = \sum_{j=0}^{t-1} \phi^{2j} V(\varepsilon_{t-j}) = \sigma^2 \sum_{j=0} \phi^{2j}$$

as $cov(\varepsilon_t, \varepsilon_{t-j}) = E(\varepsilon_t, \varepsilon_{t-j}) = 0$. Hence, from a near infinite sum of a geometric series:

$$V(y_t) = \frac{\sigma^2}{1-\phi^2} \tag{6}$$

which satisfies condition (ii) of weak stationarity.
The autocorrelation between y_t and y_{t-1} (from handout 2 (term 2), we know: $\rho(y_t, y_{t-1}) = \phi$, $\rho(y_t, y_{t-2}) = \phi^2$, $\rho(y_t, y_{t-3}) = \phi^3$ and $\rho(y_t, y_{t-j}) = \phi^j$.
This satisfies condition (iii) of weak stationarity.[1].
Consequently, the autocorrelation for the AR(1) model decays quickly to zero, for example, for $\phi = 0.8$ $\rho(y_t, y_{t-8}) = .8^8 = 0.168$, implying that the influence of any shock eventually goes to zero.



This process, which is stationary, is said to be integrated of order zero, I(0) as it needs zero differences to become stationary.

[1]This also satisfies the condition of weak dependency as $\phi^h \to 0$ as $h \to \infty$

2 Unit root I(1) processes

Consider the process for the variable y_t,

$$y_t = \mu + y_{t-1} + \varepsilon_t \tag{7}$$

where $\varepsilon_t \sim N(0, \sigma^2)$. This AR(1) model with $\phi = 1$ is known as the Random Walk (with drift for $\mu \neq 0$) model. From back substitution we have that

$$y_t = 2\mu + y_{t-2} + \varepsilon_t + \varepsilon_{t-1} \Rightarrow y_t = 3\mu + y_{t-3} + \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2}$$

$$y_t = y_0 + t\mu + \sum_{j=1}^{t} \varepsilon_j \tag{8}$$

and y_t is a function of the shock to the process, where there is no tendency for the influence of these shocks to dissipate over time (compare this to the stationary case specified in equation (4)). The element $\sum_{j=1}^{t} \varepsilon_j$ is known as the stochastic trend. Looking at equation (8):

$$E(y_t) = \mu t + y_0$$

and this violates condition (i) for a weak stationary series. (Figure 1 plots a random walk and a stationary AR(1) model as T increasees for $\mu = 0$ and $y_0 = 0$).
The variance

$$V(y_t) = E\left[(y_t - E(y_t))^2\right] = E\left[(y_t - \mu t)^2\right] = E\left[(\sum_{j=1}^{t} \varepsilon_j)^2\right] = t\sigma^2$$

and this is time varying and violates condition (ii) of a weak stationary series.
The ACF

$$cov(y_t, y_{t-1}) = E\left[\sum_{j=1}^{t} \varepsilon_j . \sum_{j=1}^{t-1} \varepsilon_j\right] = (t-1)\sigma^2$$

$$\rho_1 = \frac{(t-1)\sigma^2}{\sqrt{t\sigma^2}\sqrt{(t-1)\sigma^2}} = \frac{(t-1)}{\sqrt{t(t-1)}} = \sqrt{1 - \frac{1}{t}}$$

In general,

$$cov(y_t, y_{t-h}) = E\left[\sum_{j=1}^{t} \varepsilon_j \sum_{j=1}^{t-h} \varepsilon_j\right] = (t-h)\sigma^2$$

$$\rho_h = \frac{(t-h)}{\sqrt{t(t-h)}} = \sqrt{1 - \frac{h}{t}}$$

and as this depends on t is violates condition (iii) of weak stationarity. The ACF decays at an approximately linear rate. Note for large T the autocorrelation function stays at unity. This random walk model is said to be I(1), integrated or order 1, meaning it needs 1 difference to become stationary.

2.1 Example

Consider the stationary model:

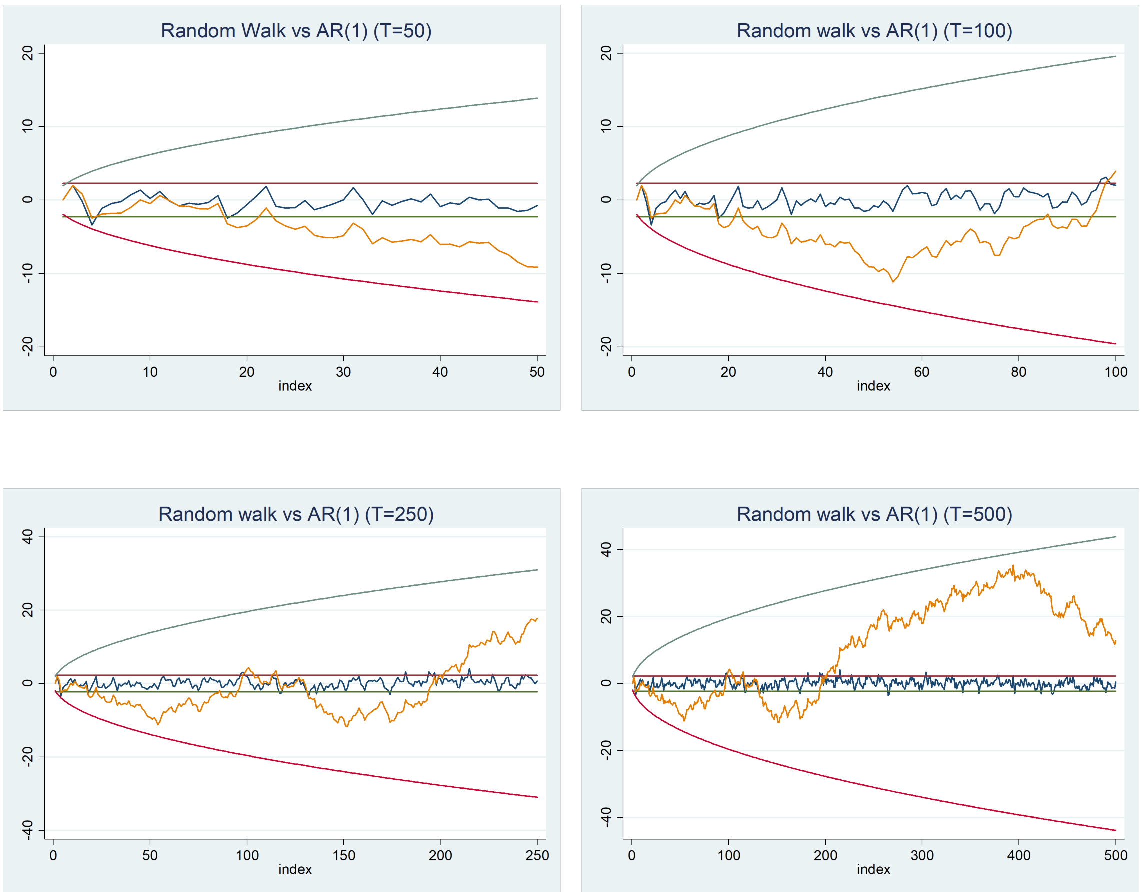
$$y_t = 0.5y_{t-1}+\varepsilon_t,\ \varepsilon_t \sim N(0,1) \Rightarrow E(y_t)=0,\ V(y_t)=\frac{1}{1-0.5^2} \Rightarrow 95\%\text{CI} = 0\pm1.96\sqrt{\frac{1}{1-0.5^2}}$$

and the non-stationary model:

$$y_t = y_{t-1}+\varepsilon_t = \sum_{j=1}^{t}\varepsilon_j,\ \varepsilon_t \sim N(0,1) \Rightarrow E(y_t)=0,\ V(y_t)=t(1) \Rightarrow \text{CI} = 0\pm 1.96\sqrt{t}$$

These are plotted in Figure 1 for different values of T

Figure 1: AR(1) vs Random Walk for Different Values of T



and while the AR(1) process bounces around a constant and this does not change, irrespective of the value of T, the random walk model exhibits increasing variance and starts to drift away from the starting point at T increases.

Figure 2 plots a stationary AR(1) series and the ACF based on $T = 250$ observations, The plots bounces around a constant and the ACF shows a geometric decay to zero.

Figure 2: Stationary AR(1) Process

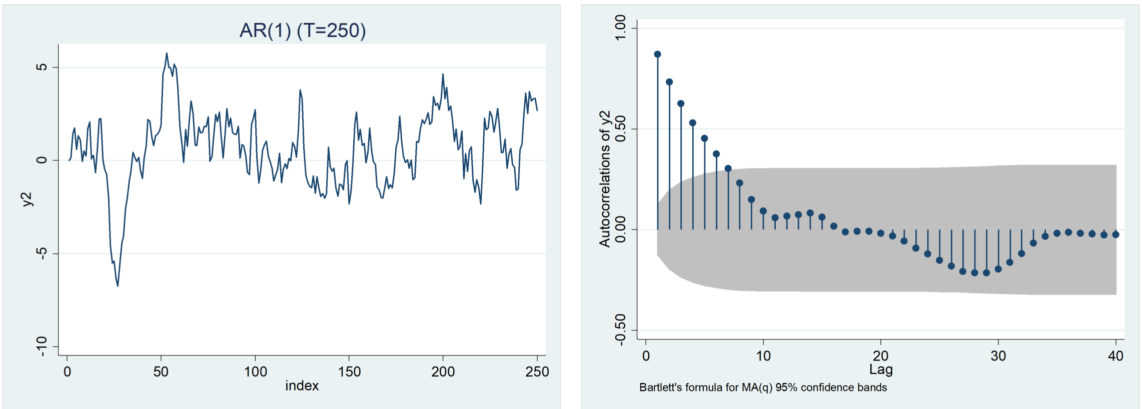


Figure 3 plots a radon walk model (without drift, $\mu = 0$) for $T = 250$, in contrast to Figure 2 here we see an apparent trend in y_t, due to the presence of the term $\sum_{j=1} \varepsilon_j$ and this is called a stochastic trend and the ACF shows a near linear decay to zero.

Figure 3: Nonstationary Random Walk Process (zero drift)

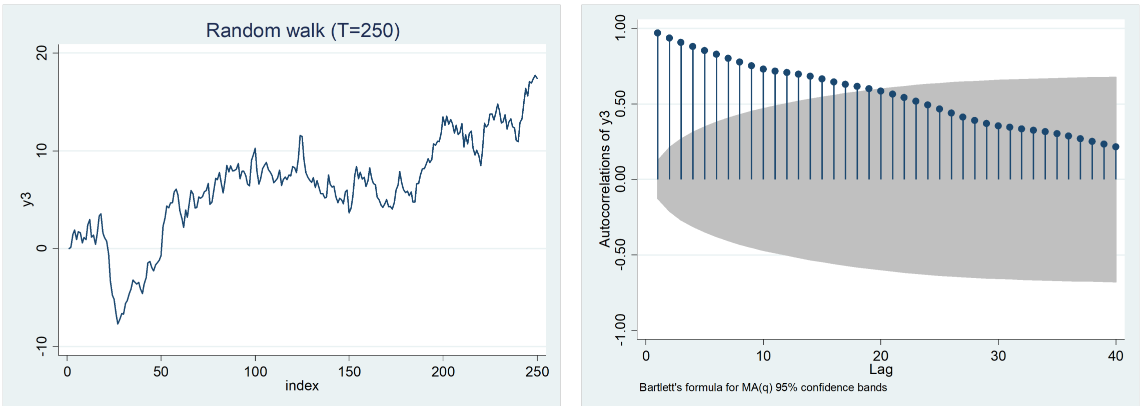
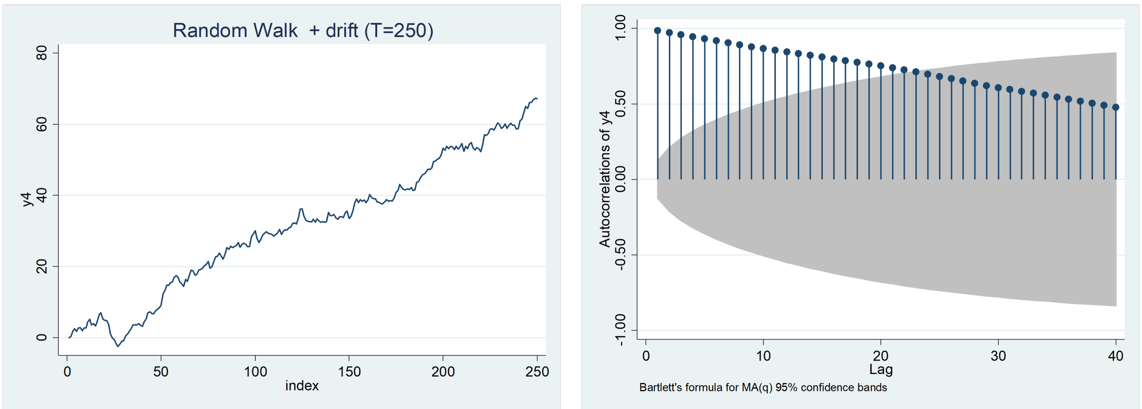


Figure 4 plots a random walk (with drift, $\mu \neq 0$) this is more trended (due to the linear growth μ term) and the ACF shows an even slower linear decay to zero (compared to Figure 3).

Figure 4: Nonstationary Random Walk Process (non-zero drift)



While limiting attention to the AR(1) and random walk model may appear restrictive, the results derived in broad principle go through for any non-stationary series. However, as an AR(1) only has 1 root is can only be either I(0) (due to the root being < 1) or I(1) (due to the root being $= 1$).

An AR(2) with 2 roots could be either I(0) (both roots are $<$1), I(1) (one root is $<$1 and one is $=$1) or I(2) (both roots are $= 1$). An I(2) process needs 2 differences to be stationary, that is, $\Delta^2 z_t = \Delta(z_t - z_{t-1}) = z_t - 2z_{t-1} + z_{t-2}$ to become stationary. In principle, an AR(p) with p roots could I(p) although it is very unlikely in economics we would see anything more than I(2).

3 Testing for Unit Roots

We will develop a test for nonstationarity for an AR(1) model before generalising the test to any form of AR model:

$$y_t = \phi y_{t-1} + \varepsilon_t \tag{9}$$

where under the null hypothesis $H_0 : \phi = 1$ we have (at least) a unit root (nonstationary random walk process), whereas under the one-sided alternative $H_1 : \phi < 1$ the model is stationary. Testing the null of stationarity in equation (9) is often rewritten by subtracting y_{t-1} from both sides as

$$y_t - y_{t-1} \equiv \Delta y_t = (\phi - 1)y_{t-1} + \varepsilon_t$$

that is:

$$\Delta y_t = \gamma y_{t-1} + \varepsilon_t \tag{10}$$

with $H_0 : \gamma = 0$ and $H_1 : \gamma < 0$.
Testing this hypothesis can be done as a t-ratio from the OLS estimation of (10) which is written as:

$$t_\gamma = \frac{\hat{\gamma} - 0}{se(\hat{\gamma})}$$

and this is known as the Dickey-Fuller test (see Dickey and Fuller (1979a)). However, this has a non-standard distribution under the null hypothesis of a unit root. In particular, the distribution is shifted to the left, such that if you compare the t-statistic to a t-distribution or N(0,1) you will reject the null hypothesis (a unit root) too frequently.
To address the non-standard nature of the distribution of the Dickey and Fuller (1979b) tabulated the critical values for some sample sizes. MacKinnon (1991) has generalised these critical values for all sample sizes, T, as:

$$CV_T = \phi_\infty + \frac{\phi_1}{T} + \frac{\phi_2}{T^2}$$

for given values of ϕ_∞, ϕ_1 and ϕ_2, such that for T=38 the 5% critical value for the no constant, no trend case as in equation is

$$-1.9393 + \frac{-0.398}{38} + \frac{-0.000}{38^2} = -1.9498$$

(cf -1.645 from a N(0,1)). For a given critical value, the null hypothesis of $H_0 : \gamma = 0$ is rejected in favour of the alternative $H_1 : \gamma < 0$ if

$$t_\gamma = \frac{\hat{\gamma} - 0}{se(\hat{\gamma})} \leq CV_T$$

.

3.1 Determining lag length

In general, the Augmented Dickey-Fuller (ADF) test for an AR(p*+1) is written as

$$\Delta y_t = \gamma y_{t-1} + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j} + \varepsilon_t \tag{11}$$

where serial correlation in the error term ε_t can be corrected by the inclusion of even more lags of Δy_t. A test for the existence of a unit root is calculated as

$$t_\gamma = \frac{\hat{\gamma} - 0}{se(\hat{\gamma})}$$

There are a variety of criteria for determining p^* in equation (11)

1. Use a variety of p^* values and see if the results of the ADF test are robust.

2. Minimise a selection criteria, e.g. Akaike Information Criteria (AIC).

3. $p^* = O(T^{1/3})$, that is, for T=125, $p^* = 5$.

4. Downward selection of p^* based on hypothesis testing

$$\begin{aligned}
p^* = 3 &\Rightarrow \Delta y_t = \gamma y_{t-1} + \delta_1 \Delta y_{t-1} + \delta_2 \Delta y_{t-2} + \delta_3 \Delta y_{t-3} + \varepsilon_t \\
p^* = 2 &\Rightarrow \Delta y_t = \gamma y_{t-1} + \delta_1 \Delta y_{t-1} + \delta_2 \Delta y_{t-2} + \varepsilon_t \\
p^* = 1 &\Rightarrow \Delta y_t = \gamma y_{t-1} + \delta_1 \Delta y_{t-1} + \varepsilon_t \\
p^* = 0 &\Rightarrow \Delta y_t = \gamma y_{t-1} + \varepsilon_t
\end{aligned}$$

In choosing between $p^* = 3$ and $p^* = 2$, test $H_0 : \delta_3 = 0$ using a standard normal distribution. If the test statistic is significant at the 5% level $p^* = 3$, otherwise $p^* \leq 2$ and you must now choose between $p^* = 2$ and $p^* = 1$ by testing $H_0 : \delta_2 = 0$. If the test statistic is significant at the 5% level $p^* = 2$, otherwise $p^* \leq 1$ and you must now choose between $p^* = 1$ and $p^* = 0$ by testing $H_0 : \delta_1 = 0$. If the test statistic is significant at the 5% level $p^* = 1$, otherwise $p^* = 0$.

NOTE:
The choice of p^* is only applicable for the unit root test, it has no implications on the lag length used in any other model you estimate.

3.2 Determining Deterministic Components

Another problem in using the ADF test revolves around the choice of the deterministic elements (i.e. whether to include a constant and/or trend or neither) in equation (11).

Equation (11) (rewritten here) is referred to as **Model A**:

$$\Delta y_t = \gamma y_{t-1} + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j} + \varepsilon_t$$

and we are testing $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$.
Under H_0 the ADF (ignoring the augmented terms $\sum_{j=1}^{p^*} \delta_j \Delta y_{t-j}$) is written:

$$\Delta y_t = \varepsilon_t$$

that is,

$$y_t = y_{t-1} + \varepsilon_t$$

and upon back substitution this can be written:

$$y_t = y_0 + \sum_{j=1}^{t} \varepsilon_j$$

in which case: $E(y_t) = y_0$.
Under H_1 we have

$$y_t = \phi y_{t-1} + \varepsilon_t$$

(where $\phi < 1$) and upon back substitution this is written:

$$y_t = \phi^{t+1} y_0 + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}$$

in which case:

$$E(y_t) = 0$$

as $\phi^{t+1} \to 0$.
To reconcile these two different expected values we require that under H_0, $y_0 = 0$ as then $E(y_t) = 0$ (under both H_0 and H_1). Given that we can never know that $y_0 = 0$, this almost rules out Model A (equation (11)) as a realistic model.
If under the null of a unit root it is unrealistic to assume $y_0 = 0$, you require **Model B** (no trend case):

$$\Delta y_t = \mu + \gamma y_{t-1} + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j} + \varepsilon_t \tag{12}$$

Under H_0 the ADF (ignoring the augmented terms $\sum_{j=1}^{p^*} \delta_j \Delta y_{t-j}$) is written:

$$\Delta y_t = \mu + \varepsilon_t$$

that is,

$$y_t = \mu + y_{t-1} + \varepsilon_t$$

and upon back substitution this ca be written:

$$y_t = y_0 + \mu t + \sum_{j=1}^{t} \varepsilon_j$$

in which case: $E(y_t) = y_0 + \mu t$.
Under H_1 we have:

$$y_t = \mu + \phi y_{t-1} + \varepsilon_t$$

(where $\phi < 1$) and upon back substitution this can be written:

$$y_t = \frac{\mu}{1-\phi} + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}$$

in which case:

$$E(y_t) = \mu/(1-\phi)$$

To reconcile these two different means we require under H_0, that $\mu = 0$ as then $E(y_t) =$ a constant (under both H_0 and H_1). Model B requires different critical values from Model A.
If under the null of a unit root it is unrealistic to assume $\mu = 0$ (under H_0), you require **Model C** (with trend case),

$$\Delta y_t = \mu + \alpha t + \gamma y_{t-1} + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j} + \varepsilon_t \tag{13}$$

Under H_0 the ADF (ignoring the augmented terms $\sum_{j=1}^{p^*} \delta_j \Delta y_{t-j}$) is written:

$$y_t = \mu + \alpha t + y_{t-1} + \varepsilon_t$$

and upon back substitution this can be written (something like):

$$y_t = y_0 + \mu t + \alpha \left[\frac{t(t+1)}{2}\right] + \sum_{j=1}^{t} \varepsilon_j$$

in which case:

$$E(y_t) = y_0 + \mu t + \alpha \left[\frac{t(t+1)}{2}\right]$$

Under H_1 we have

$$y_t = \mu + \alpha t + \phi y_{t-1} + \varepsilon_t$$

and upon back substitution this can be written:

$$y_t = \frac{\mu}{1-\phi} + \frac{\alpha t}{(1-\phi L)} + \sum_{j=0}^{t-1} \phi^j \varepsilon_{t-j}$$

in which case:

$$E(y_t) = \frac{\mu}{1-\phi} + \frac{\alpha t}{(1-\phi L)}$$

To reconcile these two different means we require under H_0, $\alpha = 0$ as then $E(y_t)$ = trend (under both H_0 and H_1). Model C requires different critical values from either Model A or Model B.

In general, most people start with model C (equation (13)) and assuming a unit root is present, that is, $\gamma = 0$, estimate the model

$$\Delta y_t = \mu + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j} + \varepsilon_t \tag{14}$$

and test whether the intercept, μ is significantly different from zero. If $\mu \neq 0$ then use model C for unit root testing, otherwise if $\mu = 0$ use model B.

3.3 Power Results for the ADF Test

The power of the ADF test is low consequently hypothesis testing is invariably done using 5% or 10% significance levels. Suppose we generate data using the model:

$$y_t = \phi y_{t-1} + \varepsilon_t, \;\; \varepsilon_t \sim N(0,1), \; t = 1, \ldots, 100$$

testing for a unit root using models B and C (model A would have strictly been the most appropriate) using $p^* = 0$ and 2 (zero is strictly appropriate)

		Model B					Model C				
		ϕ					ϕ				
Signif.	p^*	0.80	0.90	0.95	0.99	1.00	0.80	0.90	0.95	0.99	1.00
5%	0	89.0	34.6	11.7	6.30	5.10	70.2	19.6	9.60	4.80	5.70
10%	0	96.9	53.7	22.8	11.2	9.10	84.6	32.4	16.5	10.8	10.8
5%	2	69.8	26.2	11.4	6.50	5.90	47.2	17.9	7.40	6.40	5.40
10%	2	84.9	42.5	21.8	11.9	10.5	63.7	30.8	14.6	10.4	11.2

The table shows that even when Model B is used and the lag length is correct and assumed known as p^*=0, then the power of the DF test for $\phi = 0.9$ is as low as 34.6% at the 5% significance level. Including irrelevant lags and setting p^*=2, lowers the power for $\phi = 0.9$ to 26.2% at the 5% significance level. Incorrectly, estimating Model C (with a time trend) and setting p^*=2 lowers the power of the test still further, for $\phi = 0.9$ power is 17.9% at the 5%

significance level. Under specifying the lag length, p^*, of the process is equally problematic. In conclusion, it is imperative given the low power of the ADF test that the correct lag length is chosen, as well as the correct model A, B, or C. The examples section undertakes unit root tests on the 4 series.

3.4 Perron's Result

Perron (1989) and Perron (1990) show that a trend stationary series, which exhibits a structural break either in the level or the growth (trend), cannot easily be distinguished from a nonstationary process (see Figure 6); see the examples section for some empirical results of this point.

4 Alternative tests for nonstationarity

4.1 DF-GLS

This applies an Augmented Dickey–Fuller test, to a transformed time series via a generalized least squares (GLS) regression. The authors Elliott et al. (1996) have shown that this test has significantly greater power than the Augmented Dickey–Fuller test. In essence the approach you undertakes depend on whether you are testing non-stationarity against a trend stationary series, or non-stationarity against a stationary series around a constant.
In the case of testing against trend stationarity you form the transformed series:

$$\begin{aligned}
&\tilde{y}_1 = y_1 \\
&\tilde{y}_t = y_t - \alpha^* y_{t-1} \text{ for } t = 2, \dots, T \\
&x_1 = 1 \\
&x_t = 1 - \alpha^* \text{ for } t = 2, \dots, T \\
&z_1 = 1 \\
&z_t = t - \alpha^*(t-1) \text{ for } t = 2, \dots, T
\end{aligned}$$

and $\alpha^* = 1 - (13.5/T)$. You estimate the model:

$$\tilde{y}_t = \delta_0 x_t + \delta_1 z_t + \varepsilon_t$$

and formulate the residuals:

$$y_t^* = y_t - (\hat{\delta}_0 + \hat{\delta}_1 t)$$

And perform the ADF test on this transformed variable:

$$\Delta y_t^* = \alpha + \gamma y_{t-1}^* + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j}^* + \varepsilon_t$$

Testing the usual $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$, using the tabulated critical values in Cheung and Lai (1995)
NOTE:
As $T \to \infty$, $\alpha^* \to 1$ and $\tilde{y}_t = y_t - y_{t-1}$, $x_t = 0$, $z_t = 1$ in which case you are essentially just detrending the series y_t.
In the case of testing against a stationary series around a mean you form the transformed series:

$$\begin{aligned}
&\tilde{y}_1 = y_1 \\
&\tilde{y}_t = y_t - \alpha^* y_{t-1} \text{ for } t = 2, \dots, T \\
&x_1 = 1 \\
&x_t = 1 - \alpha^* \text{ for } t = 2, \dots, T
\end{aligned}$$

and $\alpha^* = 1 - (7/T)$. So e.g. for $T = 100$ the variables would be:

t	x	z
1	1	1
2	1-(1-(13.5/100))=0.135	2-(1-(13.5/100))1=1.135
3	1-(1-(13.5/100))=0.135	3-(1-(13.5/100))2=1.27
4	1-(1-(13.5/100))=0.135	4-(1-(13.5/100))3=1.405
5	1-(1-(13.5/100))=0.135	5-(1-(13.5/100))4=1.54
...	...	...
100	1-(1-(13.5/100))=0.135	100-(1-(13.5/100))99=14.5

Then estimate the model:

$$\tilde{y}_t = \delta_0 x_t + \varepsilon_t$$

And formulate the residuals:

$$y_t^* = y_t - \hat{\delta}_0$$

and perform an ADF test on the transformed variable:

$$\Delta y_t^* = \alpha + \gamma y_{t-1}^* + \sum_{j=1}^{p^*} \delta_j \Delta y_{t-j}^* + \varepsilon_t$$

testing $H_0 : \gamma = 0$ against $H_1 : \gamma < 0$, using the tabulated critical values in Cheung and Lai (1995), with the asymptotic critical values given in the table below:

	Critical Values	
	10%	5%
Constant	-1.624	-1.948
Constant + Trend	-2.550	-2.838

4.2 Kwiatkowski-Phillips-Schmidt-Shin (KPSS) Test

The KPSS (see Kwiatkowski et al. (1992)) test swaps around the null and alternative hypotheses compared to the ADF and DF-GLS test, that is, $H_0 : y_t$ is stationary (either mean stationary or trend stationary) and $H_1 : y_t$ is difference stationary (has a unit root).
The main idea of the KPSS test is that we can write y_t as

$$y_t = d_t + z_t + u_t$$

where d_t is the deterministic component (a constant OR a constant and trend term), z_t is random walk process such that

$$z_t = z_{t-1} + v_t$$

where v_t is white noise process with $E(v_t) = 0$ and $V(v_t) = \sigma_v^2$, and u_t is a stationary error (which is I(0) process but may or may not be a white noise process).
In this set up if $\sigma_v^2 = 0$ then $z_t = z_{t-1}$ and this is just a constant in which case y_t is a stationary series. If on the other hand $\sigma_v^2 > 0$ then y_t is a non-stationary process as it has within it the term z_t which is a random walk.
Therefore we are testing $H_0 : \sigma_v^2 = 0$ vs $H_1 : \sigma_v^2 > 0$ and we use a Lagrange multiplier (LM) statistic. So we undertake the test by:

1. Estimate one of the following two regressions:

$$y_t = \mu + \eta_t$$

   or

$$y_t = \mu + \alpha t + \eta_t$$

   and save the residuals $\hat{\eta}_t$

2. Construct the LM statistic

$$KPSS = \frac{\sum_{t=1}^{T} S_t^2}{T^2 \lambda}$$

   where $S_t = \sum_{j=1}^{t} \hat{\eta}_j$ and this is the partial sum of the residuals $\hat{\eta}$ [so $S_1^2 = \hat{\eta}_1^2$, $S_2^2 = (\hat{\eta}_1 + \hat{\eta}_2)^2$ $S_3^2 = (\hat{\eta}_1 + \hat{\eta}_2 + \hat{\eta}_3)^2$, ..., and $S_t^2 = (\hat{\eta}_1 + \hat{\eta}_2 + \hat{\eta}_3 + \ldots \hat{\eta}_t)^2$] and $\lambda = s^2$ is the long-run variance of the residuals $\hat{\eta}_t$.

3. KPSS test is a one-sided right tailed test: we reject H_0 at the e.g. 5% significance level if $KPSS$ is greater than 95% percentile from the asymptotic distribution, given in the table below:

	Critical Values			
	10%	5%	2.5%	1%
Constant	0.347	0.463	0.574	0.739
Constant + Trend	0.119	0.146	0.176	0.216

Appendices

A Critical values for ADF test

n	Model	MacKinnon's Critical Values Point(%)	ϕ_∞	SE	ϕ_1	ϕ_2
1	No Constant	1	-2.5658	(0.0023)	-1.960	-10.04
	no Trend	5	-1.9393	(0.0008)	-0.398	-0.00
		10	-1.6156	(0.0007)	-0.181	-0.00
1	Constant,	1	-3.4336	(0.0024)	-5.999	-29.25
	no trend	5	-2.8621	(0.0011)	-2.738	-8.36
		10	-2.5671	(0.0009)	-1.438	-4.48
1	Constant	1	-3.9638	(0.0019)	-8.353	-47.44
	+ trend	5	-3.4126	(0.0012)	-4.039	-17.83
		10	-3.1279	(0.0009)	-2.418	-7.58
2	Constant,	1	-3.9001	(0.0022)	-10.534	-30.03
	no trend	5	-3.3377	(0.0012)	-5.967	-8.98
		10	-3.0462	(0.0009)	-4.069	-5.73
2	Constant	1	-4.3266	(0.0022)	-15.531	-34.03
	+ trend	5	-3.7809	(0.0013)	-9.421	-15.06
		10	-3.4959	(0.0009)	-7.203	-4.01
3	Constant,	1	-4.2981	(0.0023)	-13.790	-46.37
	no trend	5	-3.7429	(0.0012)	-8.352	-13.41
		10	-3.4518	(0.0010)	-6.241	-2.79
3	Constant	1	-4.6676	(0.0022)	-18.492	-49.35
	+ trend	5	-4.1193	(0.0011)	-12.024	-13.13
		10	-3.8344	(0.0009)	-9.188	-4.85
4	Constant,	1	-4.6493	(0.0023)	-17.188	-59.20
	no trend	5	-4.1000	(0.0012)	-10.745	-21.57
		10	-3.8110	(0.0009)	-8.317	-5.19
4	Constant	1	-4.9695	(0.0021)	-22.504	-50.22
	+ trend	5	-4.4294	(0.0012)	-14.501	-19.54
		10	-4.1474	(0.0010)	-11.165	-9.88

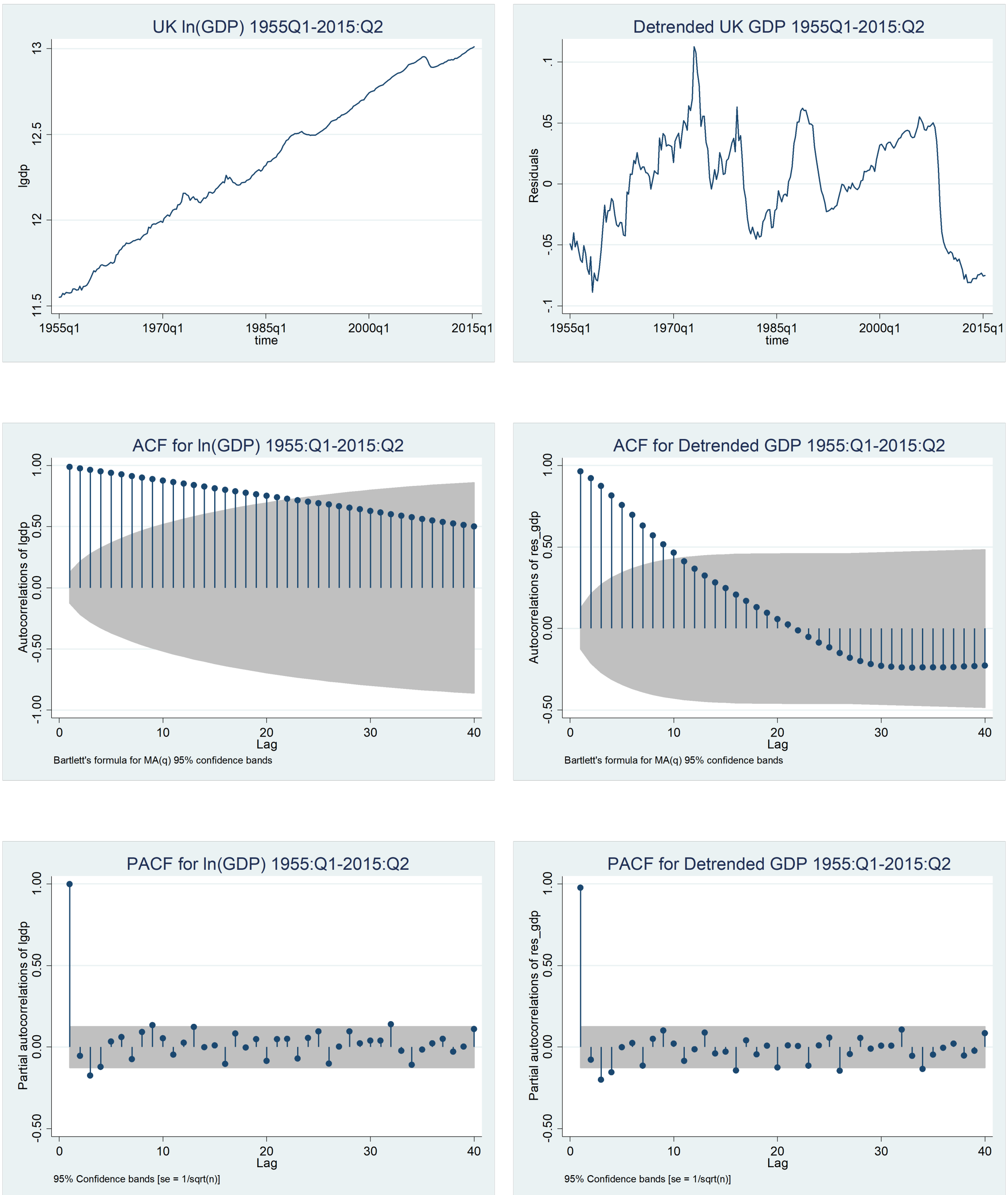
$$C(p) = \phi_\infty + \phi_1 T^{-1} + \phi_2 T^{-2}$$

Source: MacKinnon (1991). Reproduced by permission of Oxford University Press.

B Unit root test of UK real GDP

We are looking to determine the presence or otherwise of a unit root in the series for UK GDP over the period from 1955Q1-2015Q2. Fits we plot both the level and first difference of the series (measured in natural logs). See Figure A2.1 below

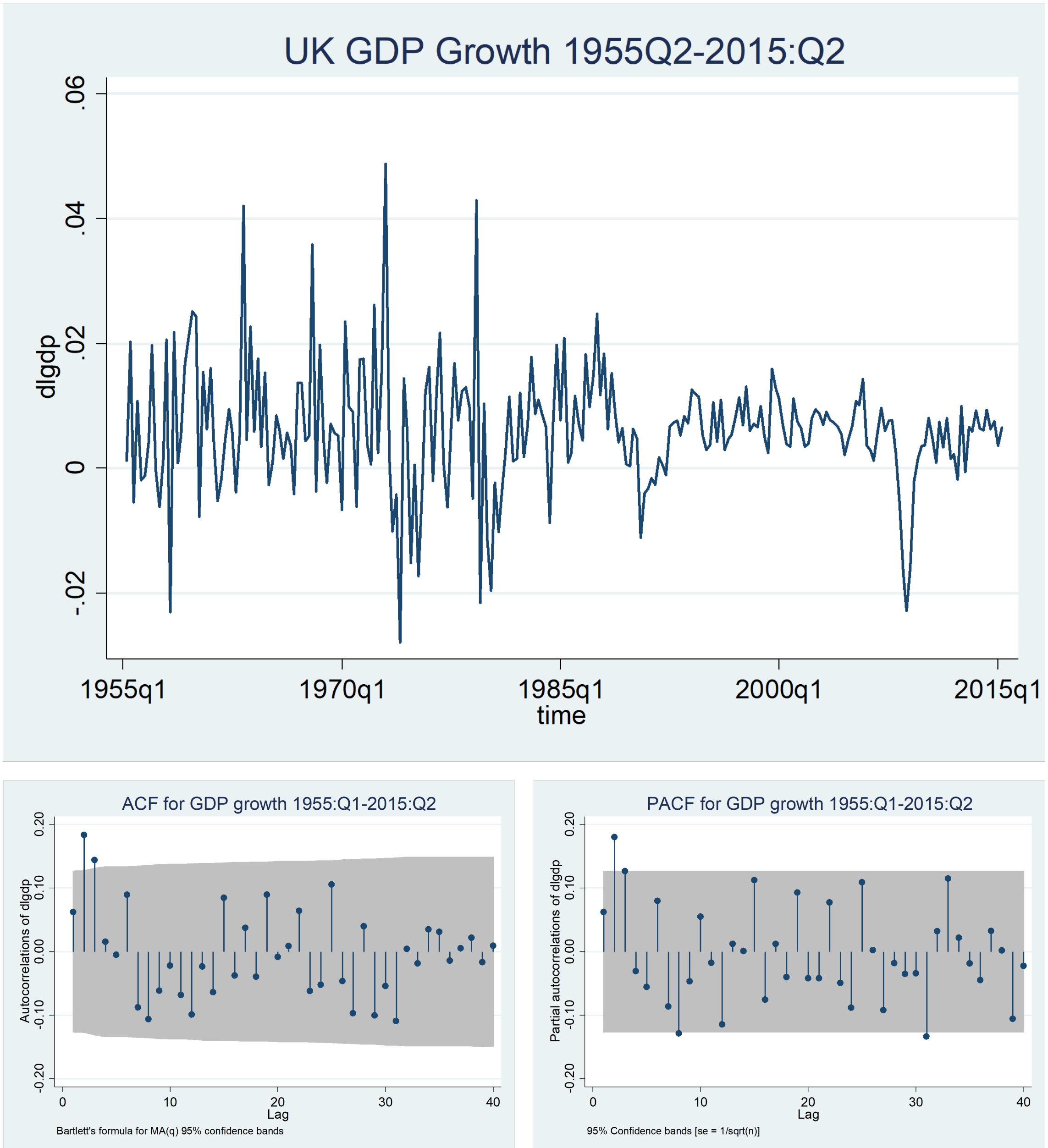
Figure A2.1 Plot, ACF and PACF of both $\ln(GDP)$ and detrended $\ln(GDP)$



Looking at the plot of LGDP and the ACF and PACF of the series, it appears to exhibit the properties of a non-stationary series. As a result consider the detrended series for LGDP

from the plot, the ACF and the PACF it still seems to look to be a very persistent series, with the 1st coefficient in the PACF close to unity and the ACF appear to be a linear rather than a geometric decay to zero. All of this suggests we look for a unit root in the series, and we use model C when undertaking out unit root test using the AF test.
Now look at the plot, ACF and PACF of the series $\Delta \ln (GDP)_t = \ln (GDP)_t - \ln (GDP)_{t-1}$ (see Figure A2.2)

Figure A2.1 Plots, ACF and PACF of $\Delta \ln(GDP)$



The ACF and PAC of the series DLGDP looks to be a stationary series, this would suggest that DLGDP is stationary, although we have a question mark over the stationarity

or otherwise of the series LGDP.
From the plot of rate of growth in GDP, which has a mean of around 0.006 (i.e. 0.6% per quarter, around 2.4% per annum), it is clear we should be using model C when undertaking out unit root test using the ADF test.
The question arises over the lag length to use for the test. With T=242, this would suggest a maximum lag length of 6. Looking at the ACF and the PACF the pattern is unclear, but it might suggest that p^*=3, might be a good starting point. Given we are using quarterly data and the previous observation I am going to try p^*=4.
So our ADF model (Model C) is: $\Delta y_t = \alpha + \mu t + \gamma y_{t-1} + \sum_{j=1}^{4} \delta_j \Delta y_{t-j} + \varepsilon_t$ And we are testing: $H_0 : \gamma = 0$ against $H_0 : \gamma < 0$, in Stata this is using the dfuller option:

```
. dfuller lgdp, trend lags(4) reg

Augmented Dickey-Fuller test for unit root         Number of obs   =       237

                               ---------- Interpolated Dickey-Fuller ---------
                  Test         1% Critical       5% Critical      10% Critical
               Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)           -2.489            -3.994            -3.432            -3.132
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.3337

------------------------------------------------------------------------------
D.lgdp       |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      lgdp |
        L1. |  -.0392018   .0157529    -2.49   0.014    -.0702402   -.0081634
        LD. |   .0511981    .065197     0.79   0.433    -.0772616    .1796579
       L2D. |   .1877513   .0647336     2.90   0.004     .0602047    .3152979
       L3D. |   .1494525   .0649659     2.30   0.022      .021448    .2774569
       L4D. |  -.0038476   .0657376    -0.06   0.953    -.1333725    .1256773
     _trend |    .000233   .0000978     2.38   0.018     .0000402    .0004258
      _cons |   .4593304   .1824119     2.52   0.012     .0999185    .8187423
------------------------------------------------------------------------------
```

The 4th lag is massively insignificant and so :

```
. dfuller lgdp, trend lags(3) reg

Augmented Dickey-Fuller test for unit root         Number of obs   =
238

                          ---------- Interpolated Dickey-Fuller ---------
               Test         1% Critical       5% Critical      10% Critical
             Statistic         Value             Value             Value
------------------------------------------------------------------------------
 Z(t)           -2.569            -3.994            -3.432            -3.132
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.2944

------------------------------------------------------------------------------
D.lgdp      |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
------------+-----------------------------------------------------------------
       lgdp |
         L1. |   -.039414   .0153422    -2.57   0.011    -.0696418   -.0091863
         LD. |   .0506579    .064368     0.79   0.432    -.0761626    .1774784
        L2D. |   .1873822   .0634569     2.95   0.003     .0623568    .3124076
        L3D. |   .1494568   .0646732     2.31   0.022      .022035    .2768786
      _trend |   .0002343   .0000953     2.46   0.015     .0000466     .000422
       _cons |   .4617731   .1777064     2.60   0.010     .1116485    .8118977
------------------------------------------------------------------------------
```

In either case the even at even the 10% critical value is that we are unable to reject the null hypothesis of a unit root.

If we test for a unit root using the DFGLS test with a maximum lag of 6 we get:

```
. dfgls lgdp, maxlag(6) trend

DF-GLS test for unit root                     Number of obs = 235
Variable: lgdp
Lag selection: User specified                 Maximum lag   =   6

                                     ———————— Critical value ————————
  [lags]      DF-GLS tau          1%            5%            10%
——————————————————————————————————————————————————————————————————————
     6          -2.223        -3.480        -2.883        -2.598
     5          -1.973        -3.480        -2.891        -2.605
     4          -2.098        -3.480        -2.898        -2.611
     3          -2.146        -3.480        -2.904        -2.617
     2          -1.791        -3.480        -2.911        -2.623
     1          -1.382        -3.480        -2.917        -2.628
——————————————————————————————————————————————————————————————————————

Opt lag (Ng-Perron seq t) = 3 with RMSE = .0095529
Min SIC  = -9.209166 at lag 2 with RMSE = .0096632
Min MAIC = -9.233938 at lag 3 with RMSE = .0095529
```

and for all lags we are unable to reject the null hypothesis.

A KPSS test [2] for the same series with a maximum lag of 6, gives the following table (including a time trend):

[2] the KPSS test needs installing using the command: ssc install kpss

```
. kpss lgdp, maxlag(6)

KPSS test for lgdp

Maxlag = 6
Autocovariances weighted by Bartlett kernel

Critical values for H0: lgdp is trend stationary

10%: 0.119   5% : 0.146  2.5%: 0.176  1% : 0.216

Lag order    Test statistic
    0           1.77
    1           .901
    2            .61
    3           .465
    4           .379
    5           .321
    6           .281
```

and we reject the null hypothesis of stationarity.

We check for a unit root in the 1st difference of the series even though we believe from the ACF and PACF that this series should be stationary. Looking at the plot of DLGDP, there is no growth in this series whatsoever and therefore we should be using model B. So our ADF model (Model b) is:

$$\Delta y_t = \alpha + \gamma y_{t-1} + \sum_{j=1}^{4} \delta_j \Delta y_{t-j} + \varepsilon_t$$

And we are testing: $H_0 : \gamma = 0$ against $H_0 : \gamma < 0$, in Stata this is using the dfuller option:

```
. dfuller dlgdp, lags(4) reg

Augmented Dickey-Fuller test for unit root         Number of obs   =
236

                            ---------- Interpolated Dickey-Fuller ---------
                 Test         1% Critical       5% Critical      10% Critical
              Statistic           Value             Value             Value
------------------------------------------------------------------------------
 Z(t)            -6.112            -3.465            -2.881            -2.571
------------------------------------------------------------------------------
MacKinnon approximate p-value for Z(t) = 0.0000

------------------------------------------------------------------------------
   D.dlgdp |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-----------+------------------------------------------------------------------
     dlgdp |
       L1. |  -.7230582   .1183016    -6.11   0.000    -.9561517   -.4899647
       LD. |  -.2337327   .1113261    -2.10   0.037    -.4530821   -.0143833
      L2D. |  -.0565037    .104239    -0.54   0.588     -.261889    .1488817
      L3D. |   .0862699   .0907766     0.95   0.343    -.0925902    .2651299
      L4D. |   .0557083    .065226     0.85   0.394    -.0728085    .1842251
           |
     _cons |   .0044061   .0009565     4.61   0.000     .0025215    .0062906
------------------------------------------------------------------------------
```

and in this case we unambiguously reject H_0 of a unit root and find that this series is stationary. This result will not be over-turned by getting rid of the insignificant terms of δ_3, δ_4 and δ_5.

C Example Non-stationary series

The figure below plots thirty I(1) series, derived as:

$$y_t^j = y_{t-1}^j + \varepsilon_t^j,\ \varepsilon_t^j \sim N(0, 25) \text{ for } t = 1, ...150$$

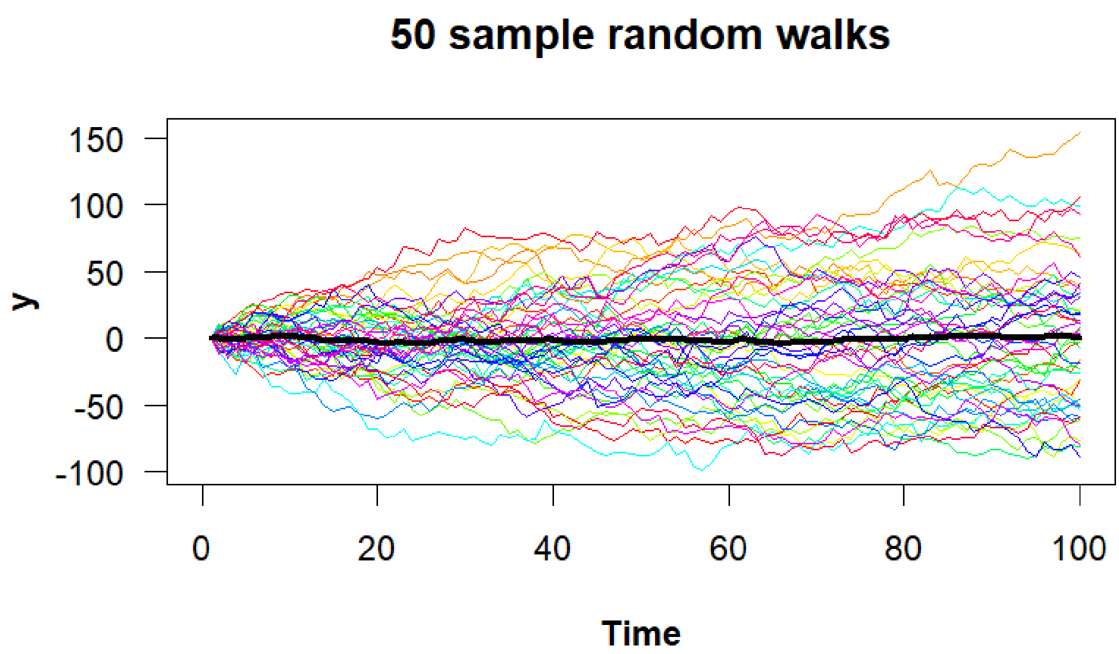
The thick black line is the average of these I(1) series showing on average, $E(y_t) = y_0 = 0$

```
clear all
global nobs = 150
global nmc = 100
set seed 10101
set obs $nobs
gen time = _n
tsset time
gen ttest=.
gen ave_y=0
forvalues i = 1/nmc {
gen y_‘i’=0
* generate errors and samples of y
replace y_‘i’ = L.y_‘i’ + rnormal(0,5) in 2/nobs
replace ave_y=ave_y+y_‘i’
\}
replace ave_y=ave_y/100
twoway /*
*/ (line y_1 time, sort) (line y_2 time) (line y_3 time) (line y_4 time) /*
*/ (line y_5 time) (line y_6 time) (line y_7 time) (line y_8 time) /*
*/ (line y_9 time) (line y_10 time) (line y_11 time) (line y_12 time) /*
*/ (line y_13 time) (line y_14 time) (line y_15 time) (line y_16 time) /*
*/ (line y_17 time) (line y_18 time) (line y_19 time) (line y_20 time) /*
*/ (line y_21 time) (line y_22 time) (line y_23 time) (line y_24 time) /*
*/ (line y_25 time) (line y_26 time) (line y_27 time) (line y_28 time) /*
*/ (line y_29  time) (line y_30 time) /*
*/ (line ave_y time, lcolor(black) lwidth(thick)), legend(off)
```

to generate something equivalent in R:

```
library(ggplot2)
library(tidyverse)
library(dplyr)
#set the seed som the graphs can be reproduced
set.seed(10101)
#setting intial value
rw_0 <- 0
# mean of the shocks (this would also be the drift parameter
m_rw <- 0
# setting the variance of the shocks
sd_rw <- 5
# setting the number of observations
n <- 100
# Generating the number of series
nseries <- 50
# Create an empty matrix for the random walk values
RW <- matrix(NA, nrow = nseries, ncol = n)
# Setting all off at intial values
RW[, 1] <- rw_0
# Generates the random walks starting at position 2
```

```
for(i in 2:n){
  RW[, i] <- RW[, i - 1] + rnorm(nseries, m_rw, sd_rw)
}
# Calculating the means of the columsn to see average values
ave_RW <- colMeans(RW)
# Create an empty plot, then add each walk as a for loop and finallly add average values
plot(NA, ylim = c(min(RW), max(RW)), xlim = c(0, n),
     xlab = "Time", ylab = "y",
     las = 1, font.lab = 2, main = "50 sample random walks")
for(i in 1:nrow(RW)){
  lines(RW[i,], col = rainbow(nseries)[i])
}
lines(ave_RW, lwd=3)
```



D Stata Commands for Non-stationarity

For quarterly data, which of the form:

	date	alc	inc
1.	1985Q1	15610	224011
2.	1985Q2	16557	228726
3.	1985Q3	16899	228942
4.	1985Q4	17218	229491
5.	1986Q1	16114	232164
6.	1986Q2	17149	233823
7.	1986Q3	17556	234862
...	...	...	

The Stata commands to declare the text date variable as quarterly data and then plot and undertake unit root testing the commands might be of the form:

```
/*To declare the series as quarterly data */
generate time = quarterly(date, "yq")
format time %tq
tsset time
/* Plot alc along with the ACF and PACF from 1990Q1 and name the graphs */
tsline alc if time>=tq(1990Q1), name(tsalc, replace)
ac alc if time>=tq(1990Q1), name(acalc, replace)
pac alc if time>=tq(1990Q1), name(pacalc, replace)
/* Test for a unit root in alc using ADF test model C and 4 lags (reporting regression) */
dfuller alc, trend regress lags(4)
/* Test for a unit root in alc using ADF test model C and 4 lags (no regression) */
dfuller alc, trend lags(4)
/* Test for a unit root in alc using an ADF test model B and 2 lags (no regression) */
dfuller alc, lags(2)
/* Test for a unit root in alc using a DF-GLS test with a maxlag of 6 + trend */
dfgls alc, maxlag(6) trend
/* Test for a unit root in alc using an DF-GLS test with a maxlag of 6 + constant */
dfgls alc, maxlag(6)
/* Test for a unit root in alc using a KPSS test with a maxlag of 6 + trend */
kpss alc, maxlag(6)
/* Test for a unit root in alc using a KPSS test with a maxlag of 6 + constant */
kpss alc, maxlag(6) notrend
/* Plot of difference in alc along with the ACF and PACF from 1990Q1 and then name the graph
tsline d.alc if time>=tq(1990Q1), name(tsdalc, replace)
ac d.alc if time>=tq(1990Q1), name(acdalc, replace)
```

```
pac d.alc if time>=tq(1990Q1), name(pacdalc, replace)
/* Test for a unit root in d.alc using an ADF test model B and 4 lags (reporting regression)
dfuller d.alc, regress lags(4)
```

E R Commands for Non-stationarity

For quarterly data, which of the form:

	date	alc	inc
1.	1985Q1	15610	224011
2.	1985Q2	16557	228726
3.	1985Q3	16899	228942
4.	1985Q4	17218	229491
5.	1986Q1	16114	232164
6.	1986Q2	17149	233823
7.	1986Q3	17556	234862
...	...	...	

The R commands to declare the text date variable as quarterly data and then plot and undertake unit root testing the commands might be of the form:

```
# In R you must install the appropriate packages (once). Then load the library associated
# with that package each time you enter R.
# I have used the following libraries over Term 2:
library(tseries)
library(urca)
library(forecast)
library(ggplot2)
library(dplyr)
library(zoo)
library(xts)
library(ggfortify)
library(dyn)
library(lmtest)
library(car)
# Read in the series as quarterly data:
uk_alc_q <- read_excel("C:/Downloads/uk_alc_q.xlsx")
uk_alc_q <- uk_alc_q %>%
  select(-c(date))
date2 <- seq(as.Date("1985-1-1"), as.Date("2015-4-1"), by = "quarters")
uk_alc_tsq <- xts(uk_alc_q, date2)
# Plot the series of alc from 1990Q1
autoplot(subset(uk_alc_tsa$alc, date2>=as.Date('1990-01-01')), geom = "line") +
  ggtitle("Plot of UK Alcohol consumtion") +
  ylab("GDP") + xlab("Year")
# Plot the ACF of alc from 1990Q1
```

```
acf_alc <- acf(subset(uk_alc_tsq$alc, date2>=as.Date('1990-1-1')))
# Plot the PACF of alc from 1990Q1
pacf_alc <- pacf(subset(uk_alc_tsq$alc, date2>=as.Date('1990-1-1')))
# Test for a unit root in alc using ADF test model C and 4 lags (reporting regression)
summary(ur.df(uk_alc_tsq$alc, type="trend", lags = 4))
# Test for a unit root in alc using an ADF test model B and 2 lags
summary(ur.df(uk_alc_tsq$alc, type="drift", lags = 2))
# Test for a unit root in alc using a DF-GLS test with a maxlag of 6 + trend
summary(ur.ers(uk_alc_tsq$alc, type= "DF-GLS", model = "trend", lag.max = 6))
# Test for a unit root in alc using an DF-GLS test with a maxlag of 6 + constant
summary(ur.ers(uk_alc_tsq$alc, type= "DF-GLS", model = "drift", lag.max = 6))
# Test for a unit root in alc using a KPSS test with a maxlag of 6 + trend
summary(ur.kpss(uk_alc_tsq$alc, type= c("tau"),  use.lag = 6))
# Test for a unit root in alc using a KPSS test with a maxlag of 6 + constant
summary(ur.kpss(uk_alc_tsq$alc, type= c("mu"),  use.lag = 6))
# Plot of difference in alc along with the ACF and PACF from 1990Q1 and then name the graphs
autoplot(subset(diff(uk_alc_tsa$alc), date2>=as.Date('1989-10-01')), geom = "line") +
  ggtitle("Plot of 1st difference in UK Alcohol consumtion") +
  ylab("d.AlC") + xlab("Year")
acf_dalc <- acf(subset(diff(uk_alc_tsa$alc), date2>=as.Date('1989-10-01')))
pacf_dalc <- pacf(subset(diff(uk_alc_tsa$alc), date2>=as.Date('1989-10-01')))
# Test for a unit root in d.alc using an ADF test model B and 4 lags (reporting regression)
summary(ur.df(diff(uk_alc_tsq$alc), type="drift", lags = 2))
```

References

Cheung, Y.-W. and Lai, K. S. (1995). Lag order and critical values of a modified dickey-fuller test. *Oxford Bulletin of Economics and Statistics*, 57(3):411–19.

Dickey, D. A. and Fuller, W. A. (1979a). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American statistical association*, 74(366a):427–431.

Dickey, D. A. and Fuller, W. A. (1979b). Distribution of the estimators for autoregressive time series with a unit root. *Journal of the American Statistical Association*, 74(366):427–431.

Dougherty, C. (2016). *Introduction to Econometrics.* OUP Catalogue. Oxford University Press.

Elliott, G., Rothenberg, T. J., and Stock, J. H. (1996). Efficient Tests for an Autoregressive Unit Root. *Econometrica*, 64(4):813–836.

Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., and Shin, Y. (1992). Testing the null hypothesis of stationarity against the alternative of a unit root : How sure are we that economic time series have a unit root? *Journal of Econometrics*, 54(1-3):159–178.

MacKinnon, J. (1991). "critical values for cointegration test. In Engle, R. F. and Granger, C. W. J., editors, *Long-Run Economic Relationships*, chapter 13. Oxford University Press, Oxford.

Perron, P. (1989). The great crash, the oil price shock, and the unit root hypothesis. *Econometrica*, 57(6):1361–1401.

Perron, P. (1990). Testing for a unit root in a time series with a changing mean. *Journal of Business & Economic Statistics*, 8(2):153–162.

Stock, J. and Watson, M. W. (2003). *Introduction to Econometrics.* Prentice Hall, New York.

Wooldridge, J. M. (2013). *Introduction to Econometrics.* Cengage.